

Calculus and Numerical Methods

1. Using the difference quotient, find $f'(-2)$ for $f(x) = -5x^2 - 2x - 5$.
2. Differentiate: $f(x) = (-4x + 1)(2x - 5)$.
3. Find the tangent line to $f(x) = 3x^2 + 4x + 3$ at $x = 3$.
4. Find and classify the critical points of $f(x) = -3x^3 + 9x + 7$.
5. Determine where $f(x) = x^3 - 48x - 4$ is increasing and decreasing.
6. Find the concavity intervals and inflection point of $f(x) = x^3 - 6x^2 + 5x - 5$.
7. A rectangle has perimeter 12. Find the dimensions that maximize its area.
8. Evaluate exactly: $\int_{-1}^1 (-3x^2 - 4x - 4) dx$.
9. Find the enclosed area between $y = -x^2 + 18$ and $y = x^2$.
10. Use a central difference with $h = \frac{1}{2}$ to estimate $f'(1)$ for $f(x) = 4x^2 - 5x - 3$.
11. Approximate $\int_0^4 (3x^2 - 4x + 3) dx$ using the composite trapezoidal rule with $n = 4$.
12. Apply composite Simpson's rule to $\int_0^4 (2x^3 - x^2 + 3x - 2) dx$ with $n = 4$.
13. Starting from $x_0 = 1$, perform 2 Newton iterations for $f(x) = x^2 - 6$.
14. Use Euler's method with $h = \frac{1}{2}$ to approximate $y(1)$ for $y' = -2y$, $y(0) = 2$.
15. Use classical RK4 with $h = \frac{1}{2}$ to approximate $y(1)$ for $y' = y$, $y(0) = 3$.

Solutions

1. Evaluate $\lim_{h \rightarrow 0} \frac{f(-2+h) - f(-2)}{h}$ to obtain 18.
2. Using the product rule, $f'(x) = -16x + 22$.
3. $f(3) = 42$ and $f'(3) = 22$, so $y - 42 = 22(x - 3)$.
4. Solve $f'(x) = -9(x^2 - 1) = 0$. The sign change gives $(-1, 1)$ minimum, $(1, 13)$ maximum.
5. Increasing on $(-\infty, -4) \cup (4, \infty)$; decreasing on $(-4, 4)$.
6. Concave up on $(2, \infty)$, down on $(-\infty, 2)$; inflection point $(2, -11)$.
7. Write $A(x) = x(6 - x)$. Its vertex occurs at $x = 3$, so the rectangle is 3×3 with maximum area 9.
8. An antiderivative is $-x^3 - 2x^2 - 4x$; evaluation gives -10 .
9. They intersect at $x = \pm 3$. Therefore $A = \int_{-3}^3 (18 - 2x^2) dx = 72$.
10. $\frac{f(\frac{3}{2}) - f(\frac{1}{2})}{2h} = 3$.
11. The tabulated values give $T_4 = 46$; absolute error 2.
12. $S_4 = \frac{368}{3}$. For this cubic, Simpson's rule is exact.
13. Using $x_{n+1} = \frac{1}{2}(x_n + \frac{6}{x_n})$; $x_1 = \frac{7}{2}$, $x_2 = \frac{73}{28}$.
14. Using $y_{n+1} = y_n + hf(t_n, y_n)$; $y_1 = 0$, $y_2 = 0$.
15. After 2 RK4 steps, $y(1) \approx 8.15204$.